Predicting 30-Day Hospital Readmissions Using Explainable Machine Learning: A Comparative Study

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***Abstract*—Hospital readmissions create significant challenges for healthcare systems. Identifying patients at risk of early readmission may support better discharge planning and follow-up care.**

This project proposes a comparative study of machine learning methods for predicting hospital readmission within 30 days using the Diabetes 130-US Hospitals dataset. We plan to compare Logistic Regression, Random Forest, XGBoost, and a Multi-Layer Perceptron (MLP) neural network.

We will investigate the effects of data preprocessing, class imbalance, and model selection on predictive performance. Explainability techniques such as SHAP will be considered to identify influential predictors. The goal is to evaluate the trade-off between predictive performance and interpretability rather than assuming that more complex models are superior.

***Index Terms*—Hospital Readmission, Machine Learning, Healthcare Analytics, XGBoost, Neural Networks, Explainable AI, SHAP**

I. Problem Definition and Motivation

Unplanned hospital readmissions can place additional pressure on healthcare resources and may indicate the need for improved post-discharge support.

Our primary research question is:

How do traditional machine learning methods and neural networks compare in predicting 30-day hospital readmissions among patients with diabetes?

We will also investigate whether explainability methods provide useful insights into model predictions.

II. Preliminary Literature Review

Strack et al. [1] investigated factors associated with hospital readmission among patients with diabetes using a large collection of hospital encounters.

Their work provides the basis for the dataset selected for this project.

Class imbalance presents an additional challenge, because positive readmission cases may represent a minority of observations. Chawla et al. [2] introduced SMOTE, a widely used technique for addressing imbalanced classification datasets.

Lundberg and Lee [3] introduced SHAP, an approach for interpreting model predictions by estimating feature contributions.

These works motivate our comparison of predictive models, imbalance-handling methods, and interpretability techniques.

III. Dataset

We plan to use the publicly available Diabetes 130-US Hospitals dataset from the UCI Machine Learning Repository.

The dataset contains approximately 101,766 hospital encounters collected between 1999 and 2008.

Potential predictors include:

- Patient age group
- Length of hospital stay
- Previous inpatient and emergency visits
- Number of diagnoses
- Medication-related variables

We will define the binary target as:

$$y = \begin{cases} 1, & \text{readmission within 30 days} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

We will examine missing values, duplicate encounters, feature availability, and class distributions before training models.

IV. Methods Under Study

We propose four supervised classification models:

1) Logistic Regression as an interpretable baseline.
2) Random Forest to capture nonlinear relationships.
3) XGBoost for gradient-boosted decision trees.
4) Multi-Layer Perceptron (MLP) as a neural-network comparison.

All models will be evaluated using a consistent data partition and preprocessing protocol.

Preprocessing will include appropriate missing-value treatment, categorical encoding, and feature scaling where required.

To reduce leakage, preprocessing will be fitted only using training data. We will also investigate whether repeated encounters from the same patient require patient-level data splitting.

V. Evaluation Methodology

Because early readmissions may be relatively uncommon, classification accuracy alone will not be considered sufficient.

We plan to report:

- Precision and Recall
- F1-score

- ROC-AUC
- Precision-Recall AUC
- Confusion matrices

Class-weighting and resampling methods may be evaluated using training data only.

For selected trained models, SHAP-based analysis will be considered to explore feature contributions.

VI. Plan and Methodology

Our proposed development phases are:

1) Data acquisition and exploratory analysis
2) Data cleaning and feature engineering
3) Baseline model development
4) Advanced machine learning experiments
5) Evaluation and error analysis
6) Model interpretation and final reporting

All experiments will be documented with reproducible Python notebooks and version-controlled source code.

VII. Team Organization

The proposed division of responsibilities is:
Bahaa Al-Juboori:
Shaaz E. Jahan:
Jason Fakidis:

VIII. Reproducibility and Repository

Project materials will be maintained in a public GitHub repository, including documentation, Python notebooks, source code, and results.
https://github.com/bahaali273/COMP9170-Hospital-Readmission

Raw data will be obtained from the original UCI source rather than redistributed through the repository.

IX. Expected Outcomes

We expect to produce a reproducible comparison of classification methods for early hospital readmission prediction.

The final study will discuss predictive performance, class imbalance, model interpretability, methodological limitations, and potential risks associated with using historical clinical data.

No experimental performance claims are made at the proposal stage.

References

[1] B. Strack et al., "Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records," *BioMed Research International*, 2014.

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[4] UCI Machine Learning Repository, "Diabetes 130-US Hospitals for Years 1999–2008." [Online]. Available: https://archive.ics.uci.edu/dataset/296/